Example 1.35

A survey was conducted among 200 magazine subscribers of three different magazines A, B and C. It was found that 75 members do not subscribe magazine A, 100 members do not subscribe magazine B, 50 members do not subscribe magazine C and 125 subscribe atleast two of the three magazines. Find

(i) Number of members who subscribe exactly two magazines.

(ii) Number of members who subscribe only one magazine.

Solution

Total number of subscribers = 200

Magazine	Do not subscribe	Subscribe
A	75	125
B	100	100
C	50	150

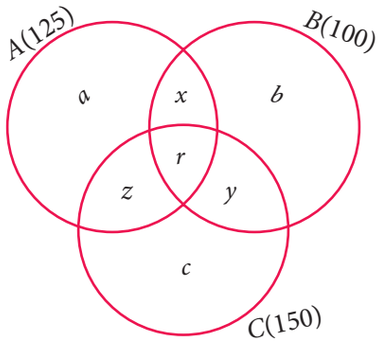


Fig.1.39

From the Venn diagram,

Number of members who subscribe only one magazine $= a + b + c$

Number of members who subscribe exactly two magazines $= x + y + z$

and 125 members subscribe atleast two magazines.

That is, $x + y + z + r = 125$ ... (1)

Now, $n(A \cup B \cup C) = 200$, $n(A) = 125$, $n(B) = 100$, $n(C) = 150$, $n(A \cap B) = x + r$

$n(B \cap C) = y + r$, $n(A \cap C) = z + r$, $n(A \cap B \cap C) = r$

We know that,

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$200 = 125+100+150-x-r-y-r-z-r+r$$

$$= 375-(x+y+z+r)-r$$

$$= 375-125-r \quad [\because x + y + z + r = 125]$$

$$200 = 250-r \quad \Rightarrow r = 50$$

From (1) $x + y + z + 50 = 125$

We get, $x + y + z = 75$

Therefore, number of members who subscribe exactly two magazines = 75.

From Venn diagram,

$$(a + b + c) + (x + y + z + r) = 200 \quad \text{... (2)}$$

substitute (1) in (2),

$$a + b + c + 125 = 200$$

$$a + b + c = 75$$

Therefore, number of members who subscribe only one magazine = 75.

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